On the Structural Impossibility of Incremental Concept Learning in Dense Weight Architectures

A Formal Analysis via Spectral Perturbation Theory

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Abstract

We prove that incremental learning of structurally new concepts is architecturally impossible in dense weight neural networks, not for algorithmic reasons, but for structural ones. The core result (Theorem 2) shows that in any associative memory with symmetric weight matrix W, no weight perturbation ΔW—regardless of rank or optimisation method—can introduce a new attractor a^* without simultaneously shifting the eigenvalues of at least one existing attractor. This follows from a compatibility condition derived from the symmetry of ΔW and the completeness of the eigenbasis of $\mathbb{R}^n$. We extend this result to Transformer attention via the equivalence between self-attention and modern Hopfield networks (Ramsauer et al., 2020), obtaining a structural conjecture for the Jacobi matrix of the attention fixed-point iteration. The analysis establishes a categorical distinction between two qualitatively different cases of incremental learning: (i) new facts about existing concepts, which require only eigenvalue shifts within the current eigenspace and are in principle tractable; and (ii) structurally new concepts, whose attractor does not yet exist in the eigenspace and whose introduction provably requires global reorganisation of all existing attractors. Catastrophic forgetting is not an engineering deficiency. It is a necessary consequence of implicit topology.

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1 Introduction

The problem of catastrophic forgetting—the tendency of neural networks to lose previously learned knowledge when trained on new data—has been studied for over three decades [5, 7]. The standard framing treats it as an engineering problem: the network overwrites old weights when exposed to new data, and the remedy is better regularisation, rehearsal, or architectural isolation.

We argue that this framing is incorrect for an important class of learning problems. There is a categorical distinction between:

1. *New facts about existing concepts*: associating a known concept with new information. This requires only weight modification within the existing eigenspace and is, in principle, tractable.
2. *Structurally new concepts*: concepts whose attractor structure does not yet exist in the eigenspace of the weight matrix. Introducing such an attractor provably requires global reorganisation—no local operation suffices.

The second case is not an engineering problem. It is a structural one. No algorithm can solve a structural problem.

This paper provides a formal proof of this claim for associative memory networks and a precisely stated conjecture for Transformer architectures.

2 Background and Related Work

2.1 Hopfield Networks and Associative Memory

The classical Hopfield network [2] stores patterns $\xi^1, \ldots, \xi^p \in \{-1, +1\}^n$ as attractors of the energy function $E(s) = -\frac{1}{2} s^\top W s$, where the weight matrix is constructed by Hebbian learning: $W = \frac{1}{n} \sum_k \xi^k (\xi^k)^\top$.

The modern Hopfield network [6] extends this to continuous states with exponential storage capacity, and—crucially for our purposes— establishes that Transformer self-attention is mathematically equivalent to one step of the modern Hopfield network update rule. This equivalence is the bridge between our results for associative memory and the conjecture for Transformers.

2.2 Catastrophic Forgetting

McCloskey and Cohen [5] identified catastrophic interference as a fundamental problem in connectionist models. Subsequent work has proposed many mitigation strategies: Elastic Weight Consolidation (EWC) [4], Progressive Neural Networks [8], and low-rank adaptation methods such as LoRA [3].

None of these approaches, however, address the structural question: *can a local weight change introduce a qualitatively new attractor without disturbing existing ones?* We show the answer is no.

2.3 Spectral Perturbation Theory

The Davis-Kahan theorem [1] provides quantitative bounds on eigenvector perturbations: for symmetric matrices W and perturbation ΔW,

$$\sin \angle(e_i, \tilde{e}_i) \leq \frac{\|\Delta W\|_2}{\delta_i},$$

where $\delta_i = \min_{j \neq i} |\lambda_i - \lambda_j|$ is the spectral gap. This establishes that any perturbation shifts all eigenvectors. Our contribution is to show that this shift is not merely quantitative but qualitatively necessary: a new attractor cannot be introduced without it.

3 Formal Framework

Definition 1 (Hopfield Network)**.** *Let $W \in \mathbb{R}^{n \times n}$ be symmetric with $W_{ii} = 0$. The energy function is $E(s) = -\frac{1}{2} s^\top W s$ for $s \in \{-1, +1\}^n$. The asynchronous update rule is $s_i \leftarrow \operatorname{sign}\left(\sum_j W_{ij} s_j\right)$.*
Definition 2 (Attractor)**.** *A pattern $\xi \in \{-1, +1\}^n$ is an attractor of W if and only if $\operatorname{sign}(W\xi) = \xi$, i.e., ξ is a fixed point of the update rule.*
Definition 3 (New Concept)**.** *A vector $a^* \in \mathbb{R}^n$ is a* structurally new concept *with respect to trained network W if $\langle a^*, \xi^k \rangle \approx 0$ for all stored attractors $\xi^1, \ldots, \xi^p$. That is, a^* lies outside the span of the stored patterns.*

4 Main Results

4.1 Attractor Approximation (Lemma)

Lemma 1 (Attractors Approximate Eigenvectors)**.** *Let $W = \frac{1}{n} \sum_{k=1}^{p} \xi^k (\xi^k)^\top$ be the Hebbian weight matrix for patterns $\xi^1, \ldots, \xi^p$. Then for each stored pattern ξ^m:*

$$W\xi^m = \xi^m + \frac{1}{n} \sum_{k \neq m} \langle \xi^k, \xi^m \rangle \xi^k.$$

When $p \ll n$ and patterns are approximately orthogonal, the cross-talk term is small, and ξ^m is an approximate eigenvector of W with eigenvalue ≈ 1.

Proof. Direct computation: $W\xi^m = \frac{1}{n} \sum_k \xi^k (\xi^k)^\top \xi^m = \frac{1}{n} \sum_k \langle \xi^k, \xi^m \rangle \xi^k = \xi^m + \frac{1}{n} \sum_{k \neq m} \langle \xi^k, \xi^m \rangle \xi^k$, where we used $\langle \xi^m, \xi^m \rangle = n$. When $p \ll n$, the patterns are approximately orthogonal by a counting argument, and the residual term has norm $O(p/\sqrt{n})$, which vanishes as $n \to \infty$ at fixed p [2]. □

4.2 The Impossibility Theorem

Theorem 2 (New Concept Requires Global Eigenvalue Shift)**.** *Let $W \in \mathbb{R}^{n \times n}$ be symmetric with eigendecomposition $W = \sum_i \lambda_i e_i e_i^\top$, where $\{e_i\}$ is an orthonormal basis of $\mathbb{R}^n$. Let $a^* \in \mathbb{R}^n$ be a structurally new concept with $\langle a^*, e_i \rangle \approx 0$ for $i = 1, \ldots, p$ (orthogonal to all stored attractors).*

Then for any symmetric perturbation ΔW that makes a^* an attractor of $W' = W + \Delta W$ (i.e., $W' a^* = \mu a^*$ for some $\mu > 0$), at least one eigenvalue of W' differs from the corresponding eigenvalue of W.

Proof. We proceed in four steps.

Step A: Necessary condition for a^* as new attractor.

If a^* is an attractor of W', then $W' a^* = \mu a^*$ for some $\mu > 0$, i.e., a^* is an eigenvector of W' with eigenvalue μ. Rearranging:

$$\Delta W \cdot a^* = (W' - W) a^* = \mu a^* - W a^* = \sum_i (\mu - \lambda_i) \langle e_i, a^* \rangle e_i =: r.$$

Note that $r \neq 0$: since a^* is a new concept not in the span of existing eigenvectors with eigenvalue μ, there exists some i for which $(\mu - \lambda_i) \langle e_i, a^* \rangle \neq 0$.

Step B: Condition for preserving old eigenvectors.

Suppose, for contradiction, that all old eigenvectors $e_1, \ldots, e_n$ remain eigenvectors of W'. Then for each j:

$$W'e_j = (\lambda_j + c_j)e_j \quad \text{for some scalar } c_j \in \mathbb{R},$$

which requires $\Delta W \cdot e_j = c_j e_j$.

Step C: Compatibility condition.

For any j, compute $\langle e_j, \Delta W \cdot a^* \rangle$ in two ways.

Using Step A: $\langle e_j, \Delta W \cdot a^* \rangle = \langle e_j, r \rangle = (\mu - \lambda_j)\langle e_j, a^* \rangle$.

Using the symmetry of ΔW and Step B: $\langle e_j, \Delta W \cdot a^* \rangle = \langle \Delta W \cdot e_j, a^* \rangle = c_j \langle e_j, a^* \rangle$.

Equating:

$$c_j \langle e_j, a^* \rangle = (\mu - \lambda_j)\langle e_j, a^* \rangle \quad \text{for all } j. \tag{$*$}$$

Step D: Case distinction and contradiction.

Consider equation $(*)$ for each j:

- **Case 1**: $\langle e_j, a^* \rangle \neq 0$. Then $c_j = \mu - \lambda_j$, so the eigenvalue of e_j shifts from λ_j to $\lambda_j + c_j = \mu$. This is the claimed eigenvalue shift.
- **Case 2**: $\langle e_j, a^* \rangle = 0$ for all j. Then a^* is orthogonal to every element of the orthonormal basis $\{e_j\}$ of $\mathbb{R}^n$, which implies $a^* = 0$. This contradicts a^* being a non-trivial new concept.

Since Case 2 is impossible, there exists at least one j with $\langle e_j, a^* \rangle \neq 0$, and the eigenvalue of e_j is shifted. This holds for *any* symmetric ΔW, regardless of rank or optimisation method. □

Corollary 3 (No Algorithm Circumvents the Result)**.** *Theorem 2 applies to rank-1, rank-k, and full-rank perturbations, and to any symmetrisation of low-rank methods (e.g., LoRA with $\Delta W = BA^\top$, symmetrised as $\frac{1}{2}(BA^\top + AB^\top)$). The result is structural, not algorithmic: no choice of learning rate, regulariser, or optimiser can avoid it.*

Proof. The proof of Theorem 2 uses only two properties of ΔW: (i) symmetry, and (ii) the completeness of $\{e_j\}$ as a basis of $\mathbb{R}^n$. Both properties hold for any symmetric perturbation of any rank. □

4.3 Distinction Between Two Learning Cases

Theorem 2 immediately yields a categorical distinction:

Corollary 4 (Two Qualitatively Different Learning Cases)**.**

1. ***New facts about existing concepts****: If the new information can be encoded as a weight shift within the current eigenspace (i.e., modifications to existing attractors), then a^* is already in* $\mathrm{span}\{e_i\}$*, and the learning is tractable in principle. Catastrophic forgetting remains a risk due to the global nature of ΔW, but is not structurally necessary.*
2. ***Structurally new concepts****: If the new concept requires an attractor a^* orthogonal to all existing eigenvectors, then by Theorem 2, its introduction shifts at least one existing eigenvalue, potentially destroying an existing attractor. This case is structurally unavoidable without full retraining.*

5 Extension to Transformer Architectures

5.1 Transformers as Modern Hopfield Networks

Ramsauer et al. [6] proved that the Transformer self-attention operation is equivalent to one update step of a modern Hopfield network with energy function:

$$F(x) = -\mathrm{lse}(xK^\top/\sqrt{d}) + \tfrac{1}{2}x^\top x + C,$$

where lse denotes the log-sum-exp function. The attention fixed point satisfies:

$$x^* = \mathrm{softmax}(x^* K^\top/\sqrt{d}) \cdot V. \qquad \text{(Self-consistency)}$$

Definition 4 (Semantic Attractor in a Transformer)**.** *Let $R \in \mathbb{R}^d$ be the residual stream at layer l for token t. We say R represents concept C if $\|R - c_C\| < \varepsilon$ for a stable vector $c_C \in \mathbb{R}^d$ satisfying the self-consistency condition above. c_C is the* semantic attractor *of concept C at layer l.*

5.2 Structural Conjecture

The Jacobi matrix of the attention fixed-point iteration at attractor c_C is:

$$J = \left.\frac{\partial f}{\partial x}\right|_{x=c_C}, \quad f(x) = \mathrm{softmax}(xK^\top/\sqrt{d}) \cdot V.$$

Stability of c_C requires $\rho(J) < 1$ (spectral radius). Semantic attractors correspond to dominant directions of J.

Conjecture 1 (Impossibility for Transformer Attractors)**.** *Let c_A, c_B be semantic attractors with $\langle c_A, c_B\rangle \approx 0$. Let a^* be a new concept such that c_{a^*} does not exist as a stable fixed point of the current attention iteration.*

Then any parameter change $\Delta\theta = (\Delta W_K, \Delta W_V, \Delta W_Q)$ that introduces c_{a^*} as a new stable attractor shifts the Jacobi matrix J of at least one existing attractor.

Evidence*: The argument of Theorem 2 applies to the linearised system via the singular value decomposition $J = U\Sigma V^\top$. By an analogous compatibility condition (replacing eigenvectors with left singular vectors and eigenvalues with singular values), introducing a^* as a new dominant singular direction of J' requires that at least one existing singular value shifts. The non-symmetry of J is handled by replacing the symmetry identity $\langle e_j, \Delta W \cdot a^*\rangle = \langle \Delta W \cdot e_j, a^*\rangle$ with the transpose identity $\langle u_j, \Delta J \cdot a^*\rangle = \langle \Delta J^\top u_j, a^*\rangle$, yielding an identical case distinction. The remaining formal step—showing that a singular value shift in J' implies attractor displacement in the nonlinear system—requires a quantitative perturbation bound and is left as an open problem.*

6 Consequences

6.1 Catastrophic Forgetting as Structural Necessity

The standard framing of catastrophic forgetting is algorithmic: the network "forgets" because gradient descent overwrites weights. Theorem 2 and Conjecture 1 reframe this:

For structurally new concepts, forgetting is not caused by a poorly chosen optimiser. It is caused by the impossibility of introducing a new eigenspace direction without globally reorganising the existing eigenstructure. The weight matrix W encodes semantic structure implicitly and globally: there is no submatrix of W that corresponds to a single concept. A new concept requires a new eigenvector, which requires shifting all others.

This explains why methods such as EWC, rehearsal, and LoRA succeed for new facts about existing concepts (Case 1 of Corollary 4) but fail for structurally new concepts (Case 2): they address the algorithmic symptom, not the structural cause.

6.2 Relation to Explicit Topology

The biological connectome avoids this limitation because its Laplacian L is sparse: a new synapse (i, j) modifies only four entries of L. By spectral perturbation theory, attractors spatially distant from (i, j) are affected only weakly. New topological paths arise locally without globally destabilising existing structure.

In contrast, a Transformer weight matrix W is dense: every gradient step shifts all singular values simultaneously, because what is local in the implicit semantic graph is global in W. This is the fundamental architectural difference between biological and artificial associative memory, and it explains why catastrophic forgetting is not an engineering deficiency but an inevitable consequence of implicit topology.

6.3 Implications for Architecture Design

Theorem 2 suggests that truly incremental concept learning requires architectures with explicit, locally modifiable topology—structures where adding a new concept corresponds to adding a local subgraph, not reorganising a global weight matrix. The biological connectome is one such architecture. Designing artificial analogues is an open engineering problem.

7 Open Problems

1. **Quantitative forgetting bound**: How large must the eigenvalue shift be for an existing attractor to collapse? Davis-Kahan provides eigenvector bounds; a complementary result bounding attractor displacement in terms of eigenvalue shift is needed.
2. **Formal proof of Conjecture 1**: The remaining step is to show that a singular value shift in the linearised system J propagates to attractor displacement in the full nonlinear attention iteration. A quantitative Lipschitz bound on the fixed-point map suffices.
3. **Threshold for Case 1 vs. Case 2**: Given a new piece of information, how do we determine whether it falls into Case 1 (tractable) or Case 2 (structurally impossible)? A geometric characterisation in terms of the angle between a^* and the current eigenspace would be useful.
4. **Architectural implications**: What is the minimal structural modification to a Transformer that enables local concept addition? Mixture-of-experts, dynamic routing, and graph-structured attention are candidate directions.

8 Conclusion

We have proved that in dense associative memory networks, no weight perturbation can introduce a structurally new concept without shifting at least one existing eigenvalue. This result is structural, not algorithmic: it holds for any symmetric ΔW of any rank, and no optimisation method can circumvent it.

The result establishes a categorical distinction between two learning cases: new facts about existing concepts (tractable in principle) and structurally new concepts (requiring global reorganisation). Catastrophic forgetting, in its most fundamental form, belongs to the second case.

For Transformer architectures, we provide a precisely stated conjecture with substantial formal evidence, leaving one quantitative bound as an open problem.

The broader implication is architectural: truly incremental concept learning requires explicit, locally modifiable topology. Solving the problem of catastrophic forgetting for structurally new concepts is not a matter of better algorithms. It requires a different kind of machine.

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